Practical 3

Dismantling and assembling of an electric geyser

AIM

1. To dismantle and assemble an electric geyser.
2. To examine the various parts of an electric geyser.
3. To study the construction and working principle of geyser
4. To find out the various faults in the geyser.

Related information

An electric geyser is an appliance used to heat water with electric supply. Generally electric geysers are available with the capacity of 5, 10, 15, 25 and 50 liters. The ratings of the geyser are ranging from 500 watts to 2000 watts.

The inner tank of the electric geyser is made up of copper with tinned coating in the inner side. In some other geyser, the tanks are made up of steel with nickel coating inside. The outer side of the geyser is made up of metal and painted to avoid rusting.

In order to protect the hotness of the water including heat loss inside the tank, a metallic cover is provided around it. Between inner and outer tank 8cm wide space is left around the tank, in which filled with glass wool. This glass wool acts as the heat insulation for the geyser.

According the capacity of the geyser, various size and type of heating elements are used. Commonly tubular type of heating element is used in geyser. The heating element is fitted inside the tube which is filled with magnesium oxide. This magnesium oxide acts as an insulator and prevents the element from touching inside the wall of the tube. The bottom head of the tube is fitted with Bakelite which is fitted with connecting pins of the heating element. The heating element, thermostat and indicating lamp are fitted in the bottom cover as in figure below.

The hot water outlet pipe is arranged in siphon manner, so that the initial water level must reach level 'A'. The water level cannot be reduced below level 'B' which ensures that the heating element will always be inside the water.

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